

Disproof of conjecture 00000000588

earthking11

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Verdict: FALSE

We disprove conjecture 00000000588 as stated. The refutation is unconditional, completely elementary, and consists of a single explicit counterexample together with the correct closed form:

$$t(\langle n, n+1, \dots, 2n-1 \rangle) = n-1 \quad (n \geq 2),$$

which equals the conjectured value $\lceil n/2 \rceil$ only at $n=2$ and $n=3$. Already at $n=4$, the smallest case not covered by the coincidence, we have

$$S = \langle 4, 5, 6, 7 \rangle, \quad \text{Ap}(S, 4) = \{0, 5, 6, 7\}, \quad t(S) = 3 \neq 2 = \lceil 4/2 \rceil.$$

The value $t=3$ is not merely computed: it is formalised in Lean 4 with a self-contained, import-free, axiom-free proof (Section 7).

1 The conjecture, quoted verbatim

From `conjectures/00000000588.md`:

English. Definition: The type $t(S)$ (the number of maximal Apéry elements). Conjecture: $t(\langle n, n+1, \dots, 2n-1 \rangle) = \lceil n/2 \rceil$ (a closed form for the type of interval semigroups).

The conjecture is filed in both English and Chinese; the Chinese original is reproduced verbatim (in Unicode) in `README.md`, and the two versions are identical in content. The PDF font does not embed CJK glyphs, so the Chinese is kept in the markdown copy rather than duplicated here.

2 Definitions and conventions

We pin every notion used by the conjecture.

Definition 1 (Numerical semigroup). A *numerical semigroup* is a submonoid $S \subseteq \mathbb{N}$ under addition with finite complement $\mathbb{N} \setminus S$. For $A \subseteq \mathbb{N}$ we write $\langle A \rangle$ for the submonoid generated by A , i.e. the set of finite sums of elements of A (including the empty sum 0).

Definition 2 (Interval semigroup). For $n \geq 1$, the *interval semigroup* is $S_n = \langle n, n+1, \dots, 2n-1 \rangle$.

Definition 3 (Multiplicity). The *multiplicity* of $S \neq \{0\}$ is its smallest positive element, $m(S) = \min(S \setminus \{0\})$.

Definition 4 (Apéry set). For a numerical semigroup S and $m \in S \setminus \{0\}$,

$$\text{Ap}(S, m) = \{s \in S : s - m \notin S\},$$

where $s - m$ is computed in $\mathbb{Z}$ (so $0 \in \text{Ap}(S, m)$ because $-m \notin S$).

The conjecture says “the Apéry elements” without naming a modulus. The standard convention, which we adopt throughout, is the *multiplicity* $m = m(S)$, the smallest positive element; for S_n that is $m = n$. Section 6 records that the verdict does not depend on this choice.

Definition 5 (Apéry order). On S put $x \leq_S y$ iff $y - x \in S$. This is a partial order on S . Its strict part is $x <_S y$ iff $x < y$ and $y - x \in S$. A *maximal Apéry element* is an element of $\text{Ap}(S, m)$ that is maximal for $\leq_S$ restricted to $\text{Ap}(S, m)$.

Definition 6 (Type). The *type* $t(S)$ is the number of maximal Apéry elements of S .

Remark 7. The maximal Apéry elements are in bijection with the pseudo-Frobenius numbers of S ; equivalently, $t(S) = |\text{PF}(S)|$ where $\text{PF}(S) = \{x \notin S : x + s \in S \ \forall s \in S \setminus \{0\}\}$. We use this only as a cross-check (Section 6).

3 The counterexample $n = 4$

Throughout this section $S = \langle 4, 5, 6, 7 \rangle$ and we use the multiplicity 4.

Lemma 8. $S = \{0\} \cup [4, \infty) = \{0\} \cup \{m \in \mathbb{N} : m \geq 4\}$.

Proof. The inclusion $\subseteq$ is clear since every generator is ≥ 4 and 0 is the empty sum. For the converse, use strong induction on $m \geq 4$. If $4 \leq m \leq 7$, then m is one of the generators 4, 5, 6, 7, hence $m \in S$. If $m \geq 8$, then $4 \in S$ and $m - 4 \geq 4$; by induction $m - 4 \in S$, so $m = 4 + (m - 4) \in S$ as S is closed under addition. $\square$

Proposition 9. $\text{Ap}(S, 4) = \{0, 5, 6, 7\}$.

Proof. By Lemma 8, $S = \{0\} \cup [4, \infty)$. Now

$$\text{Ap}(S, 4) = \{s \in S : s - 4 \notin S\}.$$

We test the elements of S :

- $s = 0$: $0 - 4 = -4 \notin S$, so $0 \in \text{Ap}(S, 4)$.
- $s = 4$: $4 - 4 = 0 \in S$, so $4 \notin \text{Ap}(S, 4)$.
- $s \in \{5, 6, 7\}$: $s - 4 \in \{1, 2, 3\}$, and $1, 2, 3 \notin S$ because S contains 0 and all integers ≥ 4 ; so $5, 6, 7 \in \text{Ap}(S, 4)$.
- $s \geq 8$: then $s - 4 \geq 4$, so $s - 4 \in S$ and $s \notin \text{Ap}(S, 4)$.

Hence $\text{Ap}(S, 4) = \{0, 5, 6, 7\}$. $\square$

Proposition 10. $t(S) = 3$, whereas $\lceil 4/2 \rceil = 2$.

Proof. We determined $\text{Ap}(S, 4) = \{0, 5, 6, 7\}$ in Proposition 9. On this set the order $\leq_S$ has strict part $x <_S y$ iff $x < y$ and $y - x \in S$. We check maximality.

The three nonzero elements are pairwise incomparable. For distinct $x, y \in \{5, 6, 7\}$ with $x < y$ we have $y - x \in \{1, 2\}$ (indeed $6 - 5 = 1$, $7 - 6 = 1$, $7 - 5 = 2$), and $1, 2 \notin S$. Hence $\neg(x <_S y)$; as $x < y$ was arbitrary, no nonzero Apéry element lies below another.

Each of 5, 6, 7 is maximal. By the previous paragraph there is no element of $\text{Ap}(S, 4)$ strictly above it, so each is maximal.

0 is not maximal. $0 <_S 5$ because $0 < 5$ and $5 - 0 = 5 \in S$.

Therefore the maximal Apéry elements are exactly $\{5, 6, 7\}$ and

$$t(S) = |\{5, 6, 7\}| = 3 \neq 2 = \lceil 4/2 \rceil. \quad \square$$

Corollary 11. *Conjecture 00000000588 is false.*

Proof. The conjecture asserts $t(S_n) = \lceil n/2 \rceil$ for the interval semigroups. For $n = 4$ Proposition 10 gives $t(S_4) = 3$ while $\lceil 4/2 \rceil = 2$; the universally quantified closed form fails at $n = 4$. $\square$

4 The true closed form

The counterexample is not an isolated anomaly; the correct closed form is $n - 1$.

Theorem 12. *For every $n \geq 2$, $S_n = \langle n, n + 1, \dots, 2n - 1 \rangle$ satisfies*

$$S_n = \{0\} \cup [n, \infty), \quad \text{Ap}(S_n, n) = \{0\} \cup \{n + 1, n + 2, \dots, 2n - 1\},$$

and every element of $\{n + 1, \dots, 2n - 1\}$ is maximal, while 0 is not. Consequently

$$t(S_n) = n - 1.$$

Proof. Step 1: $S_n = \{0\} \cup [n, \infty)$. Every generator lies in $[n, 2n - 1] \subseteq [n, \infty)$ and 0 is the empty sum, giving $\subseteq$. Conversely, let $m \geq n$. If $m \leq 2n - 1$ then m is a generator. If $m \geq 2n$, then $m - n \geq n$ and $m - n \leq m$; by strong induction on m , $m - n \in S_n$, whence $m = n + (m - n) \in S_n$.

Step 2: the Apéry set. Let $s \in S_n$. If $s = 0$, then $s - n = -n \notin S_n$, so $s \in \text{Ap}(S_n, n)$. If $n \leq s \leq 2n - 1$ and $s = n$, then $s - n = 0 \in S_n$, so $s \notin \text{Ap}(S_n, n)$; if $n + 1 \leq s \leq 2n - 1$, then $s - n \in [1, n - 1]$, and $[1, n - 1]$ contains no element of $S_n = \{0\} \cup [n, \infty)$ (because $n \geq 2$, so $1 \leq s - n \leq n - 1 < n$), hence $s \in \text{Ap}(S_n, n)$. Finally, if $s \geq 2n$, then $s - n \geq n$, so $s - n \in S_n$ by Step 1 and $s \notin \text{Ap}(S_n, n)$. Therefore $\text{Ap}(S_n, n) = \{0\} \cup \{n + 1, \dots, 2n - 1\}$.

Step 3: maximality. Let $x, y \in \text{Ap}(S_n, n)$ with $x < y$. Then $y - x \leq (2n - 1) - (n + 1) = n - 2$ and $y - x \geq 1$. Since S_n contains no integer in $[1, n - 1]$ (Step 1) and $n \geq 2$ gives $n - 2 \leq n - 1$, we get $y - x \notin S_n$, so $\neg(x <_S y)$. Hence the $n - 1$ elements $n + 1, \dots, 2n - 1$ are pairwise incomparable, and each is maximal. Moreover $0 <_S n + 1$ because $(n + 1) - 0 = n + 1 \in S_n$, so 0 is not maximal. Therefore the maximal Apéry elements are exactly $n + 1, \dots, 2n - 1$ and

$$t(S_n) = (2n - 1) - (n + 1) + 1 = n - 1. \quad \square$$

Corollary 13 (Agreement with the conjecture). *$t(S_n) = n - 1$ equals $\lceil n/2 \rceil$ if and only if $n \in \{2, 3\}$.*

Proof. For $n = 2$: $n - 1 = 1 = \lceil 2/2 \rceil$. For $n = 3$: $n - 1 = 2 = \lceil 3/2 \rceil$. For $n \geq 4$ we have $n - 1 > \lceil n/2 \rceil$: indeed $n - 1 - \lceil n/2 \rceil = \lfloor (n - 2)/2 \rfloor \geq 1$. So the two functions differ at every $n \geq 4$, and the conjecture fails for all of them. $\square$

5 The table

The following values are produced by `reproduce.py` from an exact membership test (Section 8); the computation for $n = 4$ is the one verified in Lean.

n	S_n	$\text{Ap}(S_n, n)$	$t(S_n)$	$\lceil n/2 \rceil$	agrees?
2	$\langle 2, 3 \rangle$	$\{0, 3\}$	1	1	yes
3	$\langle 3, 4, 5 \rangle$	$\{0, 4, 5\}$	2	2	yes
4	$\langle 4, 5, 6, 7 \rangle$	$\{0, 5, 6, 7\}$	3	2	no
5	$\langle 5, \dots, 9 \rangle$	$\{0, 6, 7, 8, 9\}$	4	3	no
6	$\langle 6, \dots, 11 \rangle$	$\{0, 7, \dots, 11\}$	5	3	no
7	$\langle 7, \dots, 13 \rangle$	$\{0, 8, \dots, 13\}$	6	4	no
8	$\langle 8, \dots, 15 \rangle$	$\{0, 9, \dots, 15\}$	7	4	no
9	$\langle 9, \dots, 17 \rangle$	$\{0, 10, \dots, 17\}$	8	5	no

The table is the closed form $t = n - 1$ against the conjectured $\lceil n/2 \rceil$: the two agree only at $n = 2, 3$ and separate for every $n \geq 4$. The script additionally verifies $n = 10, 11, 12$.

6 Robustness

We tried to save the conjecture by varying each convention; none succeeds.

Choice of Apéry modulus. The type is independent of the nonzero modulus used. For $n = 4$ and each $m \in \{4, 5, 6, 7, 9, 13\}$ we computed $\text{Ap}(S, m) = \{s \in S : s - m \notin S\}$ and its maximal elements:

modulus m	$\text{Ap}(S, m)$	maximal elements	count
4	$\{0, 5, 6, 7\}$	$\{5, 6, 7\}$	3
5	$\{0, 4, 6, 7, 8\}$	$\{6, 7, 8\}$	3
6	$\{0, 4, 5, 7, 8, 9\}$	$\{7, 8, 9\}$	3
7	$\{0, 4, 5, 6, 8, 9, 10\}$	$\{8, 9, 10\}$	3
9	$\{0, 4, 5, 6, 7, 8, 10, 11, 12\}$	$\{10, 11, 12\}$	3
13	$\{0, 4, \dots, 12, 14, 15, 16\}$	$\{14, 15, 16\}$	3

Every modulus gives 3 maximal elements, and the maximal set is always the translate by m of the pseudo-Frobenius set $\text{PF}(S) = \{1, 2, 3\}$: indeed $\{5, 6, 7\} - 4 = \{1, 2, 3\}$, $\{6, 7, 8\} - 5 = \{1, 2, 3\}$, and so on. Thus the type is 3 and the pseudo-Frobenius numbers are $\{1, 2, 3\}$, of size 3 as well.

Rounding conventions. For $n = 4$ all natural readings of “ $n/2$ ” give 2: $\lceil 4/2 \rceil = 2$, $\lfloor 4/2 \rfloor = 2$, $\text{round}(4/2) = 2$. Since $t = 3$, none of them restores the claim. More generally, replacing $\lceil \cdot \rceil$ by $\lfloor \cdot \rfloor$ or by rounding leaves $n = 5$ ($t = 4$, all roundings of $5/2$ are 2 or 3) as a counterexample too.

Other readings. The conjecture is a single closed-form identity for a universally quantified n ; it is not an asymptotic or an inequality, so there is no constant or error term to adjust. Either the identity $t(S_n) = \lceil n/2 \rceil$ holds for all n (it does not: $n = 4$) or the conjecture is false.

7 Formalisation

The value $t(S_4) = 3$ is formalised in the Lean 4 project under `lean4/`. `Main.lean` has **no imports at all** (not even `Std`), uses no `Mathlib` and no `sorry`; it defines

$$\begin{aligned} \text{inS } x &:= \exists a b c d, x = 4a + 5b + 6c + 7d, & \text{inAp } x &:= x = 0 \vee x = 5 \vee x = 6 \vee x = 7, \\ \text{ltS } xy &:= x < y \wedge \text{inS } (y - x), & \text{isMax } x &:= \text{inAp } x \wedge \forall y, \text{inAp } y \rightarrow \neg \text{ltS } xy, \end{aligned}$$

and proves

```
theorem not_inS_1 : ¬ inS 1    (and likewise 2, 3)
theorem not_isMax_0 : ¬ isMax 0
theorem isMax_5 : isMax 5    (and likewise 6, 7)
theorem maximals_correct :
  (∀ x, x ∈ maximals → isMax x)
  ∧ (∀ x, isMax x → x ∈ maximals)
theorem counterexample_n4 : t = 3 ∧ t ≠ 2
```

Here `maximals` is the list `[5,6,7]` and `t := maximals.length`, so `t = 3` holds by `rfl` and `t ≠ 2` by `decide`. The file also formalises $n = 3$ (`conjecture_holds_n3 : t3 = 2`) and $n = 5$ (`counterexample_n5 : t5 = 4 ∧ t5 ≠ 3`). `Check.lean` runs `#print axioms` on every theorem: the main theorems depend on *no axioms*, the rest only on the core axioms `propext` and `Quot.sound`, and none on `sorryAx`. See `lean4/README.md`.

8 Reproducibility

`reproduce.py` uses the Python 3 standard library only. For each $n = 2, \dots, 12$ it computes membership in S_n by exact dynamic programming (unbounded coin change: x is representable iff $x = 0$ or $x - g$ is representable for some generator $g \leq x$), forms $\text{Ap}(S_n, n) = \{s \in S_n : s - n \notin S_n\}$, finds the maximal elements under the Apéry order, and reports $t(S_n)$ and $\lceil n/2 \rceil$. It asserts $t(S_4) = 3 \neq 2$, that $t(S_n) = n - 1$ for every $n \geq 2$, and the robustness checks of Section 6, printing `PASS` and exiting 0 iff every check holds.

```
$ python3 reproduce.py
... (Apéry sets, maximal elements, the n = 2..12 table, robustness, all checks)
PASS: all checks verified; conjecture 00000000588 is FALSE.
```

Status against the submission rules

Rule 3 requires a LaTeX source, a PDF document, and a Lean 4 project.

- **LaTeX source** — `main.tex`, a standalone article using `geometry`, `amsmath`, `amssymb`, `amsthm`, `array`, `parskip`, `booktabs`; compiles with `tectonic main.tex`.
- **PDF** — at `build/main.pdf`, produced by `tectonic --outdir build main.tex`.

- **Lean 4 project** — `lean4/` with `lean-toolchain` pinned to `leanprover/lean4:v4.33.1`, `lakefile.toml` (library `Main`, no dependencies), `Main.lean` (no imports), `Check.lean` (`#print axioms`), and `lean4/README.md`. `lake build` exits 0; the audit reports no `sorryAx`.